

Marking Scheme

2026 · Paper 1 · 105 marks

General Rules

- M / A / PROOF mark types
- **f.t.** : 承上題錯誤但方法正確者，可獲後續步驟分。
- Deduction cap : $A(1)=1 \cdot A(2)=1 \cdot B=0$ (u, pp)
- u = 單位遺漏或錯誤；pp = 準確度不符要求。每部分最多扣 1 分。
- 答案正確且步驟合理者可獲滿分。
- 符號書寫略有出入但意思清楚者不扣分。 · Benefit of doubt
- 分數、小數或根式形式均接受，除非題目另有指明。

Section A(1) (35 marks)

1. Laws of Indices [B-]

a (3 marks)

$$\begin{aligned} \text{M1} \quad &= \frac{a^6 b^{-3}}{a^{-1} b^2} \\ \text{M1} \quad &= a^{6-(-1)} b^{-3-2} \quad \text{f.t.} \\ \text{A1} \quad &= \frac{a^7}{b^5} \quad \text{f.t.} \end{aligned}$$

2. Change of Subject [B]

a (3 marks)

$$\begin{aligned} \text{M1} \quad &Ax = 4Cx + BC \\ \text{M1} \quad &x(A - 4C) = BC \quad \text{f.t.} \\ \text{A1} \quad &x = \frac{BC}{A - 4C} \quad \text{f.t.} \end{aligned}$$

3. Algebraic Fractions [B]

a (3 marks)

$$\begin{aligned} \text{M1} \quad &= \frac{3(x+4) - 2(x-4)}{(x-4)(x+4)} \\ \text{M1} \quad &= \frac{3x+12-2x+8}{x^2-16} \quad \text{f.t.} \\ \text{A1} \quad &= \frac{x+20}{x^2-16} \quad \text{f.t.} \end{aligned}$$

4. Factorization [B+]

a (1 mark)

$$\text{A1} \quad 3(2m - 3n)$$

b (1 mark)

$$\text{A1} \quad (m - 3n)(m + 5n)$$

c (2 marks)

$$\begin{aligned} \text{M1} \quad &= (m - 3n)(m + 5n) + 3(m - 3n) \quad \text{f.t.} \\ \text{A1} \quad &= (m - 3n)(m + 5n + 3) \quad \text{f.t.} \end{aligned}$$

5. Percentages [B+]

a (4 marks)

$$\begin{aligned} \text{M1} \quad &\text{Let } female = f; male = 1.25f \\ \text{M1} \quad &1.25f + f = 180 \quad \text{f.t.} \\ \text{M1} \quad &f = 80, male = 100 \quad \text{f.t.} \\ \text{A1} \quad &difference = 20 \quad \text{f.t.} \end{aligned}$$

6. Inequalities [B+]

a (3 marks)

$$\begin{aligned} \text{M1} \quad &x > -3 \\ \text{M1} \quad &take\ union\ with\ x \leq -6 \quad \text{f.t.} \\ \text{A1} \quad &x > -3\ or\ x \leq -6 \quad \text{f.t.} \end{aligned}$$

b (1 mark)

$$\text{A1} \quad -1 \quad \text{f.t.}$$

7. Polar Coordinates [B+]

a (2 marks)

M1 $\angle AOB = 110^\circ - 50^\circ$

A1 $= 60^\circ$ ft.

b (1 mark)

A1 *equilateral, perimeter = 30* ft.

c (1 mark)

A1 **3** ft.

8. Partial Variation [B+]

a (3 marks)

M1 *Let* $f(x) = px + qx^2 : 2p + 4q = 26, 5p + 25q = 80$

M1 $p = 11, q = 1$ ft.

A1 $f(x) = x^2 + 11x$ ft.

b (2 marks)

M1 $(x + 14)(x - 3) = 0$ ft.

A1 $x = 3$ or $x = -14$ ft.

9. Statistical Tables [A-]

a (3 marks)

A1 $x = 4 + 8 = 12$

A1 $y = 22 + 13 = 35$

A1 $z = 35 + 5 = 40$

b (2 marks)

M1 *plants with height in* $[0.65, 1.25) = 10 + 13 = 23$ ft.

A1 $P = \frac{23}{40}$ ft.

Section A(2) (35 marks)

10. Locus and Circles [A]

a (2 marks)

M1 $(x - 5)^2 + (y - 7)^2 = (x - 13)^2 + (y - 1)^2$

A1 $\Gamma : 4x - 3y - 24 = 0$ ft.

b (3 marks)

M1 $H = (6, 0), K = (0, -8); HK$ is a diameter ft.

M1 $HK = 10, \text{circumference} = 10\pi \approx 31.4$ ft.

A1 $31.4 > 30, \text{so the claim is incorrect}$ ft.

11. Similar Solids [A]

a (3 marks)

M1 $\frac{V_8}{V_{12}} = \frac{8}{27}$

M1 $V_{12} \cdot \frac{19}{27} = 228\pi$ ft.

A1 $V_{12} = 324\pi \text{ cm}^3$ ft.

b (3 marks)

M1 $r = 9, \text{slant} = 15$

M1 *wet curved area* $= 135\pi \approx 424 \text{ cm}^2$ ft.

A1 $424 < 450, \text{so they disagree}$ ft.

12. Statistical Measures [A]

a (3 marks)

- M1 *median* 7.5 $\Rightarrow$ *middletwoare* 7, 8
- A1 $a = 12$ ft.
- A1 $b = 9$ ft.

b (4 marks)

- M1 (i) *add the largest distinct ages* ft.
- A1 (i) *greatest median* = 8 ft.
- M1 (ii) *add 6, 7, 8, 9; sum = 252, 34 children* ft.
- A1 (ii) *least mean* = $\frac{126}{17}$ ft.

13. Congruent Triangles [A]

a (2 marks)

- PROOF $\angle ADC = \angle AEB$; $BD = CE \Rightarrow BE = CD$ (since $BE = BD + DE$, $CD = CE + DE$); $\angle ADC = \angle AEB$ gives $AD = AE$ (isos); hence $\triangle ACD \cong \triangle AEB$
- A2 *correct proof*

b (5 marks)

- M1 (i) $AM \perp DE$, $DM = 8$
- M1 (i) $AM = \sqrt{17^2 - 8^2}$ ft.
- A1 (i) $AM = 15\text{cm}$ ft.
- M1 (ii) $AB = \sqrt{481}$, $AE = 17$, $BE = 24$ ft.
- A1 (ii) *no combination gives Pythagoras, so not a right-angled triangle* ft.

14. Polynomials [A+]

a (5 marks)

- M1 *compare* $x^4 : 2l = 6 \Rightarrow l = 3$
- M1 *compare* $x^3 : 3(m + 10) = 15 \Rightarrow m = -5$ ft.
- M1 *equal remainders* $\Rightarrow p(2) = p(-2)$
- M1 $12m + 5n = -40 \Rightarrow n = 4$ ft.
- A1 $l = 3$, $m = -5$, $n = 4$ ft.

b (5 marks)

- M1 $p(x) = 3(x^2 + 5x + 8)(2x^2 - 5x + 4)$ ft.
- M1 *disco* $x^2 + 5x + 8 = -7 < 0$ ft.
- M1 *disco* $2x^2 - 5x + 4 = -7 < 0$ ft.
- A2 *both have no real roots, so $p(x) = 0$ has 0 real roots* ft.

Section B (35 marks)

15. Permutations and Probability [A]

a (3 marks)

- M1 *5! for girls; boys in* $C_6^4 \times 4!$
- M1 $P = \frac{5! C_6^4 4!}{9!}$ ft.
- A1 $= \frac{5}{42}$ ft.

16. Standard Scores [A]

a (3 marks)

- M1 *standard deviation* = $\frac{20 - 60}{-2.5} = 16$
- M1 *Mary* = 84; *range* $\geq 84 - 20 = 64$ ft.
- A1 $64 > 59$, *so the claim is incorrect* ft.

17. Arithmetic Sequences [A]

a (2 marks)

M1 $d = \frac{555 - 666}{37}$

A1 $d = -3$ ft.

b (3 marks)

M1 $S_n = \frac{n}{2}(1335 - 3n)$ ft.

M1 $1335 - 3n > 0 \Rightarrow n < 445$ ft.

A1 $\text{greatest } n = 444$ ft.

18. Quadratic Functions [A]

a (2 marks)

M1 $f(x) = (x + 6)^2 - 157$

A1 $\text{vertex} = (-6, -157)$ ft.

b (2 marks)

M1 $\text{translate up by } 157$ ft.

A1 $g(x) = x^2 + 12x + 36$ ft.

c (2 marks)

M1 $x^2 - 12x - 121 = f(-x)$

A1 $\text{reflection about the } y\text{-axis}$ ft.

19. Trigonometry in 3-D [A+]

a (5 marks)

M1 $\frac{\sin \angle ADB}{10} = \frac{\sin 86^\circ}{15}$

M1 $\angle ADB \approx 41.7^\circ$ ft.

A1 $\angle ABD \approx 52.3^\circ$ ft.

r.t. 52.3

M1 $CD^2 = 8^2 + 15^2 - 2(8)(15) \cos 43^\circ$ ft.

A1 $CD \approx 10.7 \text{ cm}$ ft.

r.t. 10.7

b (2 marks)

M1 $\angle ABC \text{ is the angle } \iff AC \perp \text{face } BCD$

A1 $AC \text{ is not perpendicular to face } BCD, \text{ so they disagree}$ ft.

20. Centres of Triangle and Circles [A+]

a (2 marks)

PROOF PI bisects $\angle OPQ$ and PJ is part of the perpendicular bisector consideration; P, I, J collinear for the angle bisector from P to the perpendicular

A2 correct proof

b (9 marks)

M1 (i) $P = (15, 30)$

M1 (i) $\text{substitute } P, Q \text{ into } x^2 + y^2 + Dx + Ey = 0$ ft.

A1 (i) $C: x^2 + y^2 - 75x = 0$ ft.

M1 (ii) $\text{centre}(\frac{75}{2}, 0), r = \frac{75}{2}; \frac{|\frac{225}{2} + c|}{\sqrt{10}} = \frac{75}{2}$ ft.

M1 (ii) $c = -\frac{225}{2} \pm \frac{75\sqrt{10}}{2}$ ft.

M1 (ii) $STUV = \frac{|c_1^2 - c_2^2|}{6}$ ft.

M1 (ii) simplify ft.

M1 (ii) $STUV = 2812.5\sqrt{10} \approx 8894$ ft.

A1 (ii) $8894 < 9000, \text{ so the claim is incorrect}$ ft.